Title : E-Commerce Shopping Cart Using a Doubly Linked List in C++

Code :

```
#include <iostream>
#include <string>

using namespace std;

class Node {
public:
    int productId;
    string productName;
    double price;
    int quantity;
    Node* next;
    Node(int id, string name, double p, int q) {
        productId = id;
        productName = name;
        price = p;
        quantity = q;
        next = nullptr;
    }
};
class ShoppingCart {
private:
    Node* head;
public:
    ShoppingCart() {
        head = nullptr;
    }
```

```
// 1. Add item to the cart
void addItem(int id, string name, double price, int quantity) {
    Node* newNode = new Node(id, name, price, quantity);

    if (head == nullptr) {
        head = newNode;
    } else {
        Node* temp = head;
        while (temp->next != nullptr) {
            temp = temp->next;
        }
        temp->next = newNode;
    }
    cout << "Item added to cart.\n";
}

// 2. Update quantity
void updateQuantity(int id, int newQuantity) {
    Node* temp = head;
    while (temp != nullptr) {
        if (temp->productId == id) {
            temp->quantity = newQuantity;
            cout << "Quantity updated.\n";
            return;
        }
        temp = temp->next;
    }
    cout << "Product not found in cart.\n";
}
```

```
// 3. Remove item
void removeItem(int id) {
    if (head == nullptr) {
        cout << "Cart is empty.\n";
        return;
    }

    // If the item to remove is the first node
    if (head->productId == id) {
        Node* temp = head;
        head = head->next;
        delete temp;
        cout << "Item removed.\n";
        return;
    }

    Node* current = head;
    Node* previous = nullptr;

    while (current != nullptr && current->productId != id) {
        previous = current;
        current = current->next;
    }

    if (current != nullptr) {
        previous->next = current->next;
        delete current;
        cout << "Item removed.\n";
    } else {
        cout << "Product not found.\n";
    }
}
```

```
    // 4. Simple display function
    void displayCart() {
        if (head == nullptr) {
            cout << "Cart is empty.\n";
            return;
        }

        Node* temp = head;
        double totalCost = 0.0;

        cout << "\n--- Cart Details ---\n";
        while (temp != nullptr) {
            cout << "ID: " << temp->productId
                 << " | Name: " << temp->productName
                 << " | Price: Rs" << temp->price
                 << " | Qty: " << temp->quantity << "\n";

            totalCost += (temp->price * temp->quantity);
            temp = temp->next;
        }
        cout << "Total Cost: Rs" << totalCost << "\n\n";
    }
};

int main() {
    ShoppingCart cart;
    int choice, id, qty;
    double price;
    string name;
```

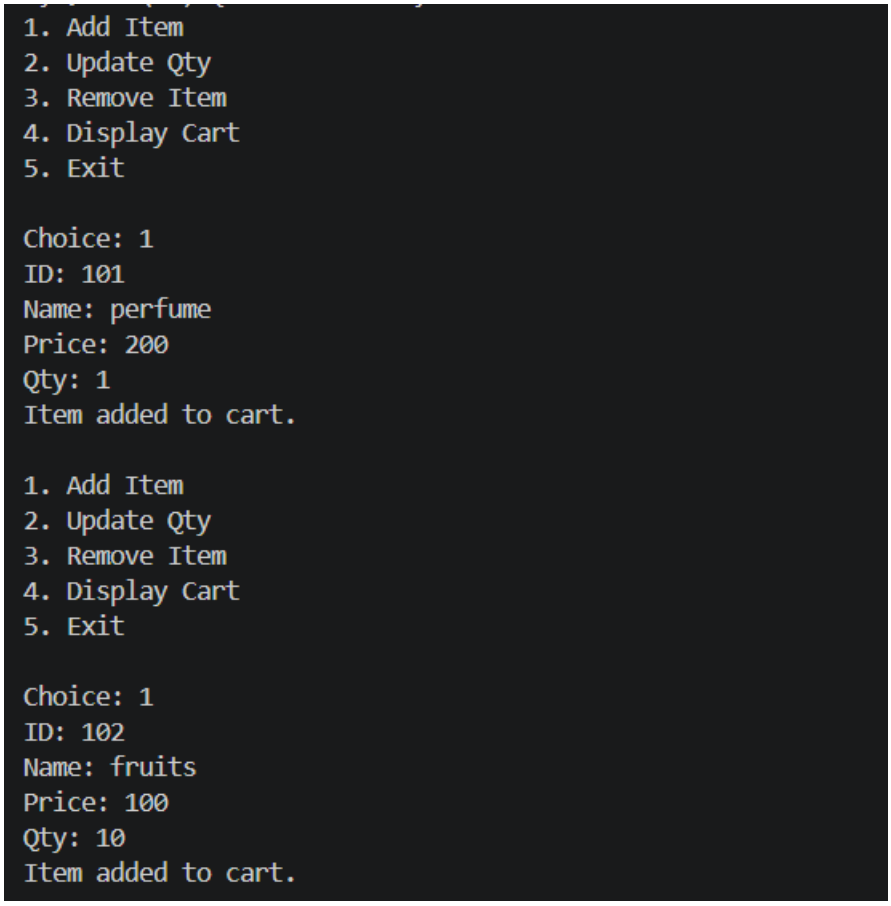
```
do {
    cout<<"1. Add Item"<<endl;
    cout<<"2. Update Qty"<<endl;
    cout<<"3. Remove Item"<<endl;
    cout<<"4. Display Cart"<<endl;
    cout<<"5. Exit\n"<<endl;
    cout << "Choice: ";
    cin >> choice;

    switch (choice) {
        case 1:
            cout << "ID: "; cin >> id;
            cout << "Name: "; cin >> name;
            cout << "Price: "; cin >> price;
            cout << "Qty: "; cin >> qty;
            cart.addItem(id, name, price, qty);
            break;
        case 2:
            cout << "ID: ";
            cin >> id;
            cout << "New Qty: ";
            cin >> qty;
            cart.updateQuantity(id, qty);
            break;
        case 3:
            cout << "ID: ";
            cin >> id;
            cart.removeItem(id);
            break;
        case 4:
            cart.displayCart();
            break;
```

```
            case 5:
                cout << "Exiting...\n";
                break;
            default:
                cout << "Invalid choice.\n";
        }
        cout << "\n";
    } while (choice != 5);

    return 0;
}
```

Output :



```
ID: 101
Name: perfume
Price: 200
Qty: 1
Item added to cart.

1. Add Item
2. Update Qty
3. Remove Item
4. Display Cart
5. Exit

Choice: 1
ID: 102
Name: fruits
Price: 100
Qty: 10
Item added to cart.

1. Add Item
2. Update Qty
3. Remove Item
4. Display Cart
5. Exit
```

```
Choice: 4

--- Cart Details ---
ID: 101 | Name: perfume | Price: Rs200 | Qty: 1
ID: 102 | Name: fruits | Price: Rs100 | Qty: 10
Total Cost: Rs1200


1. Add Item
2. Update Qty
3. Remove Item
4. Display Cart
5. Exit

Choice: 3
ID: 101
Item removed.

1. Add Item
2. Update Qty
3. Remove Item
4. Display Cart
5. Exit
```

```
Choice: 4

--- Cart Details ---
ID: 102 | Name: fruits | Price: Rs100 | Qty: 10
Total Cost: Rs1000


1. Add Item
2. Update Qty
3. Remove Item
4. Display Cart
5. Exit

Choice:
```

```
Choice: 2
ID: 102
New Qty: 12
Quantity updated.

1. Add Item
2. Update Qty
3. Remove Item
4. Display Cart
5. Exit

Choice: 4

--- Cart Details ---
ID: 102 | Name: fruits | Price: Rs100 | Qty: 12
Total Cost: Rs1200


1. Add Item
2. Update Qty
3. Remove Item
4. Display Cart
5. Exit
```